

0.1 Mathematical Operations

When we combine two or more numbers to create a new number, we perform a **mathematical operation**. **Addition**, **subtraction**, **multiplication**, and **division** are our fundamental mathematical operations, and we call them the four basic **arithmetic operations**. When we use these operations, we create **calculations**. The value we get from performing a mathematical operation is often called the **answer** to the calculation.

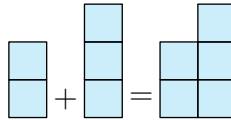
0.2 Addition

Addition with Quantities: Adding

When we have a quantity and want to add more, we use the **plus sign**

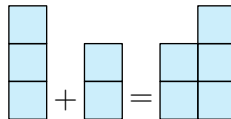
+. If we have 2 and want to add 3, we write

$$2 + 3 = 5$$



The order in which we add the numbers does not matter; starting with 2 and then adding 3 is the same as starting with 3 and then adding 2:

$$3 + 2 = 5$$



The language box

An addition problem consists of two or more **terms** and one **sum**. In the addition problem

$$2 + 3 = 5$$

both 2 and 3 are terms, while 5 is the sum.

Common ways to say $2 + 3$ are:

- "2 plus 3"
- "2 added to 3"
- "2 and 3 added together"

Adding numbers is also called **summing**.

0.1 Addition is Commutative

The sum is the same regardless of the order of the terms.

Example

$$2 + 5 = 7 = 5 + 2$$

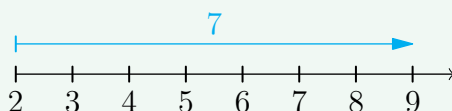
$$6 + 3 = 9 = 3 + 6$$

Addition on the Number Line: Moving to the Right

On a number line, addition with positive numbers means moving to the right:

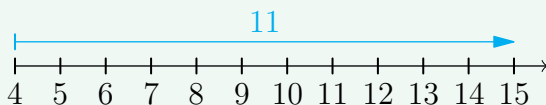
Example 1

$$2 + 7 = 9$$



Example 2

$$4 + 11 = 15$$



A New Meaning of =

$+$ gives us the ability to express numbers in many different ways. For example, $5 = 2 + 3$ and $5 = 1 + 4$. In this context, $=$ means "has the same value as". This also applies to subtraction, multiplication, and division, which we will cover in the next three sections.

0.3 Subtraction

Subtraction with Quantities: Taking Away

When we have a quantity and remove part of it, we use the **minus sign** $-$. If we have 5 and want to remove 3, we write

$$5 - 3 = 2$$



The language box

A subtraction problem consists of two or more **terms** and one **difference**. In the subtraction problem

$$5 - 3 = 2$$

both 5 and 3 are terms, while 2 is the difference.

Common ways to say $5 - 3$ are:

- "5 minus 3"
- "5 subtracted by 3"
- "3 taken from 5"
- "3 subtracted from 5"

A New Interpretation of 0

In chapter [Chapter ??](#), we mentioned that 0 can be interpreted as "nothing". Subtraction gives us the ability to express 0 using other numbers. For example, $7 - 7 = 0$ and $19 - 19 = 0$.

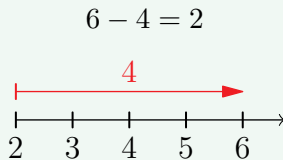
Subtraction on the Number Line: Moving to the Left

In [section 0.2](#), we saw that $+$ (with positive numbers) means moving *to the right* along the number line. With $-$ we do the opposite; we move *to the left*!! ¹ :

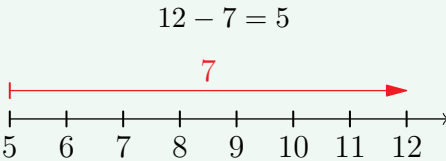
Note

In *Example 1* and *Example 2* below, we move in the opposite direction of the arrow. This means we start at the arrowhead and move to the end. This may seem a bit odd at first, but it will be useful, especially in [Chapter ??](#), to think this way.

Example 1



Example 2



¹In figures with number lines, red arrows indicate that you start at the arrowhead and move to the other end.

Inverse Operations

The figures showing addition and subtraction on the number line illustrate why we say that addition and subtraction are **inverse operations**.

For example: Since $2 + 4 = 6$, we know that $6 - 4 = 2$ and that $6 - 2 = 4$.

Canceling Terms

If we have an expression where numbers with the same value are added as many times as they are subtracted, we say they **cancel** each other out. In the expression

$$4 + 3 - 4 + 4 + 7 - 4 + 1$$

we have as many 4s added as we have 4s subtracted. The total sum and difference of the 4s is equal to 0. Thus, we can simplify our expression to

$$3 + 7 + 1$$

Note! In [Chapter ??](#), the word *cancellation* will have a different meaning.

0.4 Multiplication

Multiplication with Whole Numbers: Introductory Definition

When we add the same numbers together, we can use the **multiplication sign** $\cdot$ to write our calculations more concisely:

Example

$$4 + 4 + 4 = 4 \cdot 3$$

$$8 + 8 = 8 \cdot 2$$

$$1 + 1 + 1 + 1 + 1 = 1 \cdot 5$$

The language box

A multiplication problem consists of two or more **factors** and one **product**. In the multiplication problem

$$4 \cdot 3 = 12$$

4 and 3 are factors, while 12 is the product.

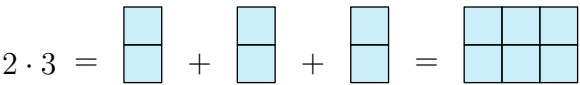
Common ways to say $4 \cdot 3$ are:

- "4 times 3"
- "4 multiplied by 3"
- "4 multiplied with 3"

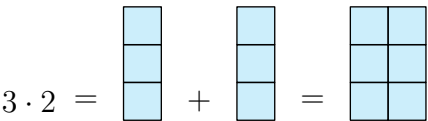
Many websites and books in English use the symbol $\times$ as the multiplication sign. In most programming languages, the symbol $*$ is used.

Multiplication of Quantities

Let’s use a figure to visualize the multiplication problem $2 \cdot 3$:



And then we can observe the product of $3 \cdot 2$:



0.2 Multiplication is Commutative

The product is the same regardless of the order of the factors.

Example

$$3 \cdot 4 = 12 = 4 \cdot 3$$

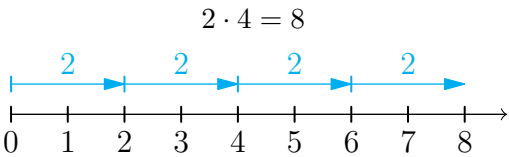
$$6 \cdot 7 = 42 = 7 \cdot 6$$

$$8 \cdot 9 = 72 = 9 \cdot 8$$

Multiplication on the Number Line

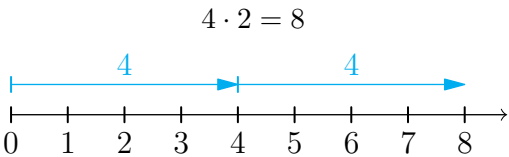
We can also use the number line to solve multiplication problems. For example, we can find what $2 \cdot 4$ is by thinking like this:

” $2 \cdot 4$ means moving 2 spaces *to the right*, 4 times.”



The number line can also be used to see that the order of the factors does not matter:

” $4 \cdot 2$ means moving 4 spaces *to the right*, 2 times.”



Final Definition of Multiplication with Positive Whole Numbers

It might seem most natural to interpret "2 times 3" as "3, 2 times". Then

$$\text{"2 times 3"} = 3 + 3$$

On page 8, we presented $2 \cdot 3$, i.e., "2 times 3", as $2 + 2 + 2$!. With this interpretation, $3 + 3$ corresponds to $3 \cdot 2$, but the fact that multiplication is a commutative operation ([Rule 0.2](#)) means that one interpretation does not exclude the other; $2 \cdot 3 = 2 + 2 + 2$ and $2 \cdot 3 = 3 + 3$ are two expressions with the same value. Thus, we can also write $2 \cdot 3 = 3 + 3$ and $3 \cdot 2 = 2 + 2 + 2$.

0.3 Multiplication as Repeated Addition

Multiplication with a positive whole number can be expressed as repeated addition.

Example 1

$$4 + 4 + 4 = 4 \cdot 3 = 3 + 3 + 3 + 3$$

$$8 + 8 = 8 \cdot 2 = 2 + 2 + 2 + 2 + 2 + 2 + 2$$

$$1 + 1 + 1 + 1 + 1 = 1 \cdot 5 = 5$$

Note

The fact that multiplication with positive whole numbers can be expressed as repeated addition does not exclude other expressions. It is not wrong to write that $2 \cdot 3 = 1 + 5$.

0.5 Division

`:` is the **division sign**. Division has three different meanings, here shown with $12 : 3$:

0.4 The Three Meanings of Division

- **Equal Distribution of Quantities**
 $12 : 3 =$ "The number in each group when 12 is divided into 3 equal groups"
- **Number of Times One Number Fits into Another**
 $12 : 3 =$ "How many times 3 fits into 12"
- **Inverse Operation of Multiplication**
 $12 : 3 =$ "The number you multiply 3 by to get 12"

The language box

A division problem consists of a **dividend**, a **divisor**, and a **quotient**. In the division problem

$$12 : 3 = 4$$

12 is the dividend, 3 is the divisor, and 4 is the quotient.

Common ways to say $12 : 3$ are:

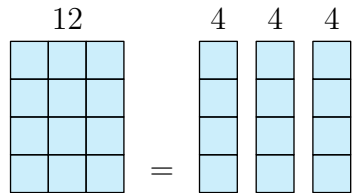
- "12 divided by 3"
- "12 divided into 3"
- "12 over 3"

In some contexts, $12 : 3$ is called the "**ratio** between 12 and 3". Then, 4 is the **ratio value**.

Often, `/` is used instead of `:`, especially in programming languages.

Equal Distribution of Quantities

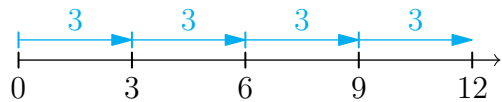
The calculation $12 : 3$ tells us to distribute 12 into 3 equal groups:



We see that each group contains 4 squares, which means that

$$12 : 3 = 4$$

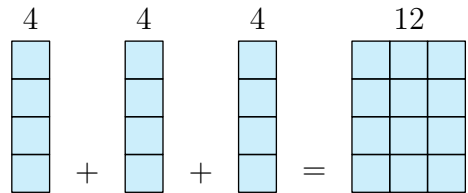
Number of Times One Number Fits into Another



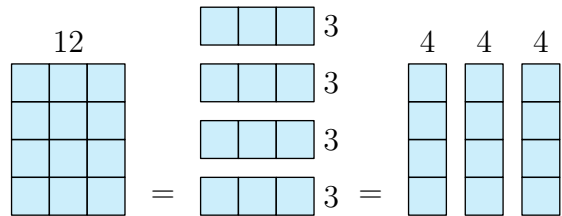
3 fits 4 times into 12, so $12 : 3 = 4$.

Inverse Operation of Multiplication

We have seen that if we divide 12 into 3 equal groups, we get 4 in each group. So, $12 : 3 = 4$. If we add these groups together again, we naturally get 12:



But this is the same as multiplying 4 by 3. So, if we know that $4 \cdot 3 = 12$, we also know that $12 : 3 = 4$. Additionally, we then know that $12 : 4 = 3$.



Example 1

Since $6 \cdot 3 = 18$, then

$$18 : 6 = 3$$

$$18 : 3 = 6$$

Example 2

Since $5 \cdot 7 = 35$, then

$$35 : 5 = 7$$

$$35 : 7 = 5$$